

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 4

For the tutorial in Week 5 on 15 February, let us discuss

- Ex. 6.5.1, 6.5.4, 6.5.12, 6.5.11, 6.5.14, 6.5.19, 6.5.32, 8.1.10, additional exercises.

Ex. 6.5.11. How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where x_i , $i = 1, 2, 3, 4, 5$, is a non-negative integer, such that

- (a) $x_1 \geq 1$?
- (b) $x_i \geq 2$ for $i = 1, 2, 3, 4, 5$?
- (c) $0 \leq x_1 \leq 10$?
- (d) $0 \leq x_1 \leq 3$ and $x_2 \geq 15$?

Ex. 6.5.14. How many solutions are there to the inequality

$$x_1 + x_2 + x_3 \leq 11,$$

where x_1, x_2, x_3 are non-negative integers?

[Hint: introduce an auxiliary variable x_4 such that $x_1 + x_2 + x_3 + x_4 = 11$.]

Ex. 6.5.19. How many positive integers strictly lesser than 1,000,000 have the sum of their digits equal to 19 ?

Ex. 6.5.1, 6.5.4, additional exercise.

- (a) In how many different ways can five elements be selected **in order** from a set with three elements when repetition is allowed ?
- (b) How many ways are there to select five unordered elements from a set with three elements when repetition is allowed ?
- (c) How many ways are there to assign three distinct jobs to five (of course distinct) employees if each employee can be given more than one jobs ?

Ex. 6.5.12, additional exercise.

- (a) How many strings of 10 ternary digits (0, 1, or 2) are there that contain exactly two 0's, three 1's, and five 2's ?
- (b) Let n be a multiple of k . How many ways can n books be placed on k distinguishable shelves if no two books are the same, the positions of the books on the shelves do not matter, and each shelf has exactly n/k books?

Ex. 6.5.32 modified. How many ways can n books be placed on k distinguishable shelves

- (a) if the books are indistinguishable copies of the same title?
- (b) if no two books are the same, and the positions of the books on the shelves matter?

Ex. 8.1.10. A binary string is a string containing only 0's and 1's.

- (a) find a recurrence relation for the number of bit strings of length n that contain two consecutive 0's.
- (b) what are the initial conditions ?
- (c) find the answer when $n = 4$.

Additional Exercise on coffee. Bottle T has 4 litres of tea and bottle C has 4 litres of coffee. In each step you pour 1 litre of liquid from bottle T into bottle C, stir, pour back 1 litre of liquid from bottle C into bottle T, and stir again. (This is a recipe for Kopi Cham, see https://en.wikipedia.org/wiki/Kopi_tiam) Let $f(n)$ be the amount of coffee in bottle T after n steps.

- (a) Calculate the values $f(0)$, $f(1)$, and $f(2)$.
- (b) Write down the recurrence formula for $f(n)$.
- (c) Solve the recurrence from part (2).
- (d) What is the limiting value of $f(n)$ as n approaches infinity?

Hints.

Ex. 6.5.32(b). The number of ways can be computed in a two step process. Step 1: Count how many ways there are to distribute the n books into k shelves as if the n books are indistinguishable and the order on the shelves do not matter. Step 2: Count how many permutations there are of the n books, this is to take care of the fact that the books are distinguishable and the order of the books on the shelves matter. Finally, apply the product rule.

Ex. 8.1.10(a). Just like the lecture examples, consider the last bit. For the case when the last bit is 0, we split into two cases, whether the last two bits are 10 or 00.

Additional Exercise on coffee This question is not difficult, just long. Note that when there is $f(n)$ amount of coffee in bottle T, it means there is $4 - f(n)$ amount of coffee in bottle C.